F continuation: background nuisance estimation

22 October 2026 revision

This manuscript companion reports a bounded exploratory continuation; it does not replace the historical results in the paper. Existing credits and licence scopes remain unchanged. This editorial revision is dated 22 October 2026. Scientific execution occurred on 19 October 2026 UTC (20 October in New Zealand at completion); no new scientific computation accompanies this revision.

Question and four procedures

The question is whether background nuisance treatment changes a fixed finite inhibition procedure, and whether conditional likelihood compensation or numerical stopping sensitivity helps explain the observed differences. No physical or causal explanation follows merely from a favourable contrast. All alternatives estimate inhibition η; the null fixes $\eta = 0$.

Cell	Free null coordinates	Free alternative coordinates
I_{00}	none	η
I_{10}	a	(a, η)
I_{01}	d	(d, η)
I_{11}	(a, d)	(a, d, η)

Here a, d are background log level and trend; other parameters are supplied according to the pinned law. Supplied/supplied is not a fixed-η oracle. The original procedure is distinct from the separately labelled solver and geometry diagnostics.

Strict marked-history accounting

For a fixed observed history, write

$$\lambda(t) = \exp\{a + dx(t) - \eta G(t)\} + nH(t), \qquad \ell = \sum_i \log \lambda(t_i^-) + \sum_i \log f_\beta(q_i) - \int_0^T \lambda(t)\, dt.$$

Here $x(t) = 2t/(24394\,\text{days}) - 1$ is the actual affine observation coordinate, with t measured in days; it is not $1/t$ or a Taylor approximation. Only earlier events enter $G(t_i^-), H(t_i^-)$. The current mark affects later history, not its own event intensity. Initial quiet and terminal exposure, all earlier parents and the exact supplied projected/paired clock convention are retained. No clock or mark repair is introduced.

The direct contrast retains event and complete compensator accounting. With the same supplied n, trigger kernel/productivity and mark law, trigger exposure and common mark terms cancel between null and alternative. This is an explicit cancellation, not a reason to omit those terms in comparisons that change them. Event-log terms cannot be additively allocated to latent causes. The primary numerical procedure is order 16 with mandatory order-32 and other prescribed controls. A finite candidate does not override an unresolved optimizer or control status. Bounded recipes establish neither global optima nor equal effective complexity.

Conditional amplitude profile

Fix d, η, n, kernels, mark law and history, and let

$$A = e^a, \quad u_i = e^{dx_i - \eta G_i}, \quad v_i = nH_i, \quad Q = \int_0^T e^{dx(t) - \eta G(t)} dt, \quad C = n \int_0^T H(t) dt.$$

The ground negative log likelihood, omitting only the common mark constant, is $L(A) = AQ - \sum_i \log(Au_i + v_i) + C$. Hence

$$L_A = Q - \sum_i \frac{u_i}{Au_i + v_i}, \qquad L_{AA} = \sum_i \left(\frac{u_i}{Au_i + v_i} \right)^2 \geq 0.$$

Convexity holds in A, not generally in a. With $p_i = Au_i/(Au_i + v_i)$, $L_{aa} = AQ - \sum_i p_i(1 - p_i)$ can be negative. The constrained profile uses $A \in [e^{-12}, 1]$: a nonnegative lower-end L_A selects the lower boundary; a nonpositive upper-end L_A selects the upper boundary; otherwise a bracketed scalar root is used. When every $v_i = 0$, N/Q gives the unconstrained stationary point, restricted to the allowed interval; an empty history selects the lower boundary.

The implementation reports actual derivative/KKT tolerance, amplitude bracket width, stopping reason and exact/approximate status. Its parameter named `a_tolerance` gates width in A, not absolute error in a. Interior acceptance can be curvature/bracket-width limited; it must not be described as meeting a fixed derivative threshold alone. The conditional mesh fixes the remaining parameters and is not a confidence region. The $\eta = 0$ analytic background branch and positive-η quadrature branch have a separately recorded numerical seam. All four prescribed difference steps are retained, with truncation/cancellation and boundary-domain cautions; no best step repairs a scientific acceptance rule.

Realized-event geometry

At a fixed point the event log-intensity columns are

$$c_i = (p_i,\ p_i x_i,\ -p_i G_i), \qquad Z = W^{1/2} C_{\text{event}}.$$

Main06 uses $W = I_N$, so every actual event weight is one. The factors p_i already enter the event columns; quadrature weights are separate. The raw Gram matrix is $Z^\top Z$. Nonzero weighted columns are normalized by their Euclidean norms before numerical SVD/projection. If $\widetilde{z}_\eta$ is the scaled inhibition column and $P_{a,d}$ the numerical least-squares projection on the scaled nuisance columns, the reported residual is

$$r_\eta = \frac{\|(I - P_{a,d})\widetilde{z}_\eta\|_2^2}{\|\widetilde{z}_\eta\|_2^2}.$$

This is a *squared* residual fraction, not a norm ratio. A zero inhibition column is explicitly undefined for interpretation, despite its reported zero convention. The event Gram excludes compensator derivatives and random-history variation: it is not a full observed Hessian, ensemble Fisher information, sampling covariance, global identification theorem or causal quantity. Low residuals describe conditional path overlap only.

Completed portions and missing work

The independently reviewed geometry stage is partial: six of 11 cases completed, two timed out, one was started-unknown at the tree-RSS stop and two were unattempted. All 11 remain in accounting

and figures. The saved portion has 253 of 396 solver rows and 840 of 1,540 profile points, with 840 corresponding order-32 controls. Two old failure witnesses were intentionally selected separately from nine metadata extrema; neither selection estimates population prevalence or detection probability.

The old witnesses reproduced maxls=40 ABNORMAL returns; diagnostic maxls=100 returned success at unchanged objective and essentially unchanged parameters. This is stopping-flag sensitivity, not proof of the precise old cause, gradient convergence or a corrected production outcome. No old unresolved status is promoted, and maxls=100 is not substituted into calibration.

The separate ended nine-case full-procedure tail run produced seven complete cases and 31 of 36 controlled method outcomes. One finite candidate lacked final controls and four methods were absent; all five remain unknown. The large alternative case timed out during its Powell control, while the largest null case failed the unchanged internal 128 MiB feature-cache limit. An external 2 GiB tree lease did not relax that internal limit or repair the saved failure. These stress cases do not demonstrate complete bank coverage.

Calibration and uncertainty contract

Let R be the planned reference count, K the controlled reference indices, $U = R - |K|$, and $E_q = \sum_{r \in K} \mathbf{1}\{T_r \geq T_q\}$ for a controlled query statistic. The add-one rank interval is

$$p_q \in \left[\frac{1 + E_q}{R + 1}, \frac{1 + E_q + U}{R + 1} \right].$$

Ties use $\geq$. An uncontrolled query remains unknown. An upper endpoint at most α gives a definite rejection; a lower endpoint above α gives a definite non-rejection; otherwise the decision remains unknown. For N planned queries, k definite rejections and u unknowns give accounting bounds $[k/N, (k+u)/N]$, not sampling confidence intervals.

For the paired nuisance interaction,

$$D_i = I_i^{00} - I_i^{10} - I_i^{01} + I_i^{11}, \qquad \overline{D} = N^{-1} \sum_i D_i, \qquad \text{MCSE} = \sqrt{\frac{\sum_i (D_i - \overline{D})^2}{N(N-1)}}.$$

The declared point summary requires all planned paired rows resolved and $N \geq 2$; it is conditional on the fixed reference bank, not a survivor-only estimate or reference-bank uncertainty measure.

The executed method used 2,000 shared-reference and paired-query resamples as a separately gated descriptive sensitivity analysis, not automatic confirmatory, simultaneous or unconditional coverage. Historical independence is source-reported, and adaptive pilot/cost-query reuse remains relevant. Both production jobs ended before final calibration. Independent review reconstructed base decisions, replayed all 2,000 shared-reference plans and checked all 80 chunk hashes. It independently recalculated eight declared query-replicate witnesses. A separate audit checked all 120 saved endpoint quantiles. This does not mean every resampled query statistic was independently recalculated.

Actual bounded checkpoint

The final inventory retains 2,149 histories: 999 references, 1,000 null queries and 150 alternative queries. Of 8,596 planned method outcomes, 7,068 have controlled statistics and 1,528 remain unknown. There are 1,752 histories with all four controlled, 18 partial, three with exceptions and 376 never attempted. One of the partial claimed histories was interrupted with an uncontrolled final method. Registration is not complete computation. The two production phases produced 1,639 new terminal records and reused 133 immutable carried records; the claim count was 1,640.

At $\alpha = 0.05$, the final accounting is:

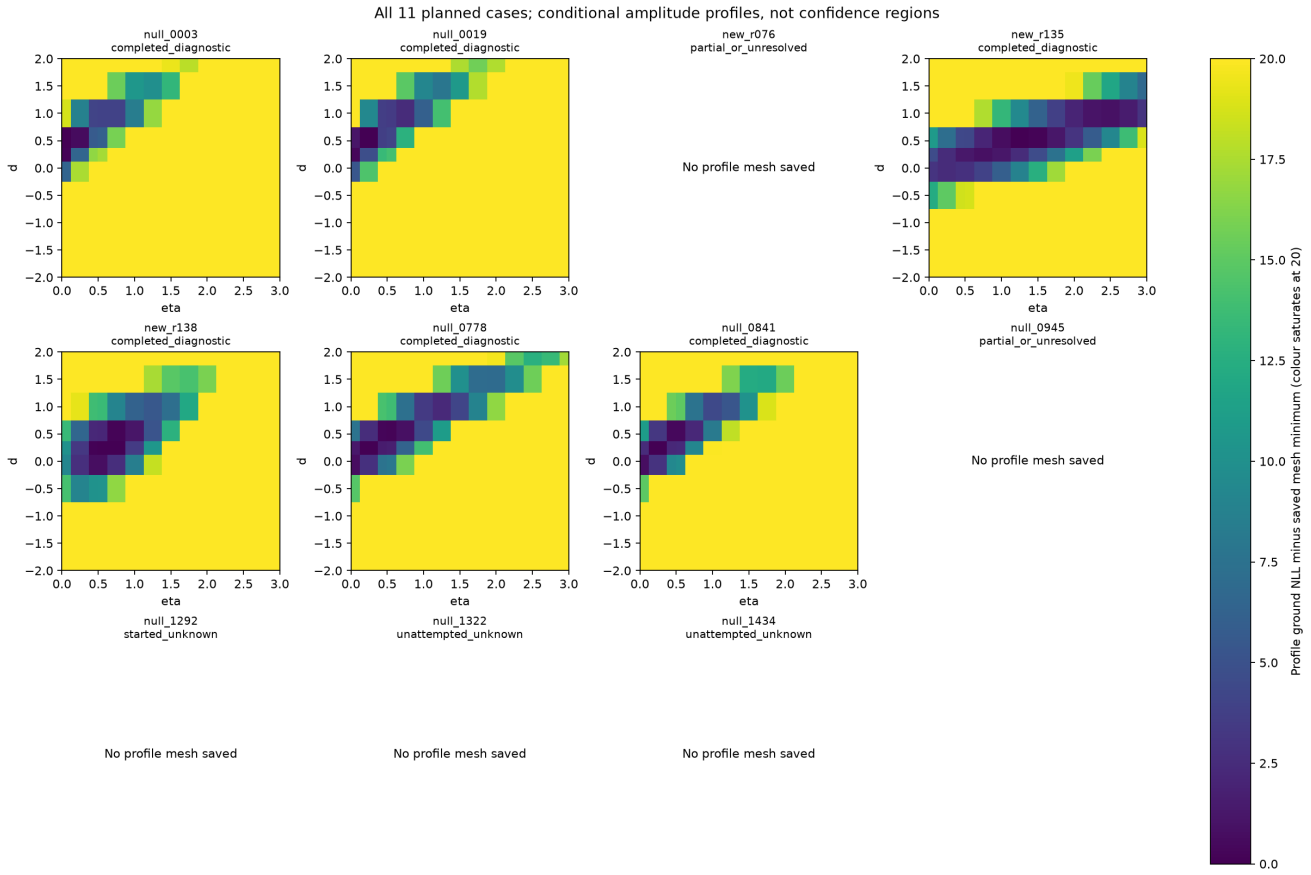


Figure 1: All 11 planned geometry cases, including explicit missing panels. Conditional profile NLL is not a confidence region; colour saturation is labelled in the reviewed image.

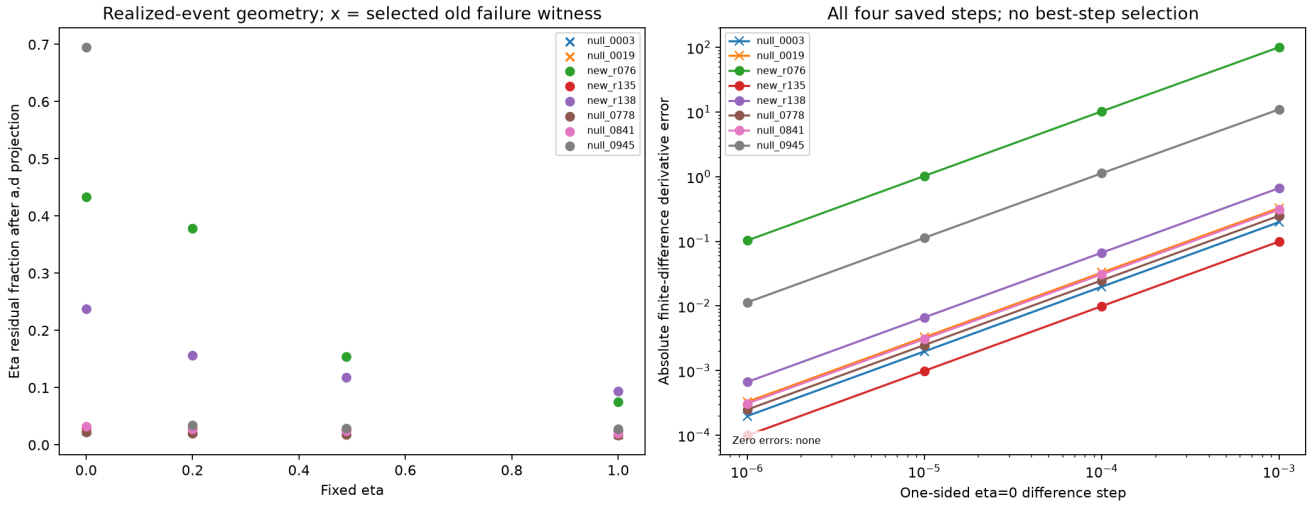


Figure 2: Realized-event overlap and all saved $\eta = 0$ derivative steps. Positive absolute errors use a log display; exact zeros, if any, are listed separately without floors, clipping or imputation. Selected old witnesses remain distinct. Neither panel establishes a causal mechanism.

Queries	Cell	Reject	Non-reject	Unknown	Rejection bounds
Null ($N = 1000$)	I_{00}	30	593	377	$[0.030, 0.407]$
	I_{10}	35	588	377	$[0.035, 0.412]$
	I_{01}	31	586	383	$[0.031, 0.414]$
	I_{11}	31	589	380	$[0.031, 0.411]$
Alternative ($N = 150$)	I_{00}	150	0	0	$[1, 1]$
	I_{10}	122	28	0	$[0.8133, 0.8133]$
	I_{01}	150	0	0	$[1, 1]$
	I_{11}	65	82	3	$[0.4333, 0.4533]$

The finite fitting recipe reported here gives 122 rejections out of 150 for I_{10} and 65 rejections, 82 non-rejections and three unknown decisions for I_{11}. These are distinct from the paper's historical counts of 123/150 and 68/150, respectively. A separate source-reported metadata check matched historical and continuation input identities and case roles; this compact packet does not reproduce it. No paired historical/continuation outcome comparison is reported; the counts should not be combined or interpreted as a paired change in detection.

These are bounds over all planned queries, not confidence intervals or estimates conditional on successful fitting. Controlled reference counts are 998, 998, 992 and 997 in cell order; controlled null-query counts are 623, 623, 619 and 620; alternative counts are 150, 150, 150 and 148. Thus a controlled query can still have an unresolved decision because reference uncertainty straddles the threshold. Null missingness depends on history cost, traversal and deadlines; the broad bounds do not establish empirical 5% size.

The nuisance-interaction accounting bounds are $[-0.765, 0.752]$ under the null and $[-0.38, -0.36]$ under the alternative. The required full-pair point mean and MCSE are unavailable for both roles. No survivor-only point estimate is substituted. On these 150 saved alternative histories, supplying level and estimating trend gives 150 definite rejections; estimating both nuisance coordinates is associated with materially fewer detections. The signed interaction is a procedure contrast, not a causal effect or proof of physical inhibition.

Completed descriptive resampling

All 2,000 declared replicates completed, using one shared reference plan across the four cells and both roles, and separate paired query plans by role. Lower and upper accounting endpoints are summarized separately with linear/type-7 quantiles. For the alternative I_{11} cell, the 2.5%, median and 97.5% quantiles of the lower endpoint are $(0.3800, 0.4333, 0.4667)$ for bank-only resampling and $(0.3400, 0.4333, 0.5200)$ when paired queries are also resampled. The corresponding upper-endpoint quantiles are $(0.4067, 0.4533, 0.4800)$ and $(0.3600, 0.4533, 0.5400)$. These summarize sensitivity of incomplete accounting, not certified coverage. All 120 endpoint quantiles and counts of every replicate status are retained in the JSON/CSV tables; individual statuses remain in the saved chunk JSON. No unresolved replicate is discarded. In the alternative paired-query mode, 126 replicates have a resolved point interaction and 1,874 remain unresolved. This does not resolve the original full-pair point.

Posthoc accounting of the continuation pilot

This subsection concerns the earlier 25-query pilot of this continuation, not the paper's separate η-planning experiment (6/25 versus 11/25). After inspecting the aggregate calibration, a separate saved-evidence check fixed the original 25 alternative IDs and compared their unchanged carried scores under the old and expanded reference banks. This was descriptive posthoc accounting: no fit, rank plan or

history was regenerated. For I_{10}, the original cohort changed from $(6, 11, 8)$ to $(19, 6, 0)$ in reject/non-reject/unknown counts. The remaining 125 alternatives gave $(103, 22, 0)$ under the same final bank. For I_{11}, the original cohort changed from $(6, 17, 2)$ to $(8, 17, 0)$ and the remaining cohort gave $(57, 65, 3)$. Both cohorts have all rejections in I_{00} and I_{01}. The same 25 fitted scores yield different decisions under the new reference calibration. This comparison does not isolate bank size from reference composition, rank resolution or missingness. Neither this comparison nor the additional query cohort establishes population causality, independent validation or nominal test size.

Next decision

The continuation stops at this incomplete evidence checkpoint. Incomplete null coverage prevents a calibrated-size conclusion. Reopening the study would first require a bounded demonstration of a substantially cheaper, numerically valid recovery procedure, with consistent rules across roles and cells. Old failures would remain preserved, and any reference-bank change would require recalibration of the entire contrast. No such recovery, cache change, fresh bank or wider scientific campaign accompanies this manuscript revision.

Availability and bounded interpretation

The 2,149 reused bank byte/clock identities were reverified; the historical RNG and generating process were not replayed. The private NPZ bank is not included in source-only material. Exact excluded inputs, settings and receipts are required for historical reproduction; constructed fixtures are separate. The accompanying saved summary, accounting tables and figures document the reported numerical checkpoint; they do not replace the omitted raw inputs.

Conditional overlap, solver flags and controlled calibration outcomes must be interpreted together with failures and missing tails. They do not identify physical inhibition or justify an automatic cache change, retry, new bank, broader search or next cycle. Any such work remains a separately reviewed proposal. This companion makes no claims about the other research tracks.

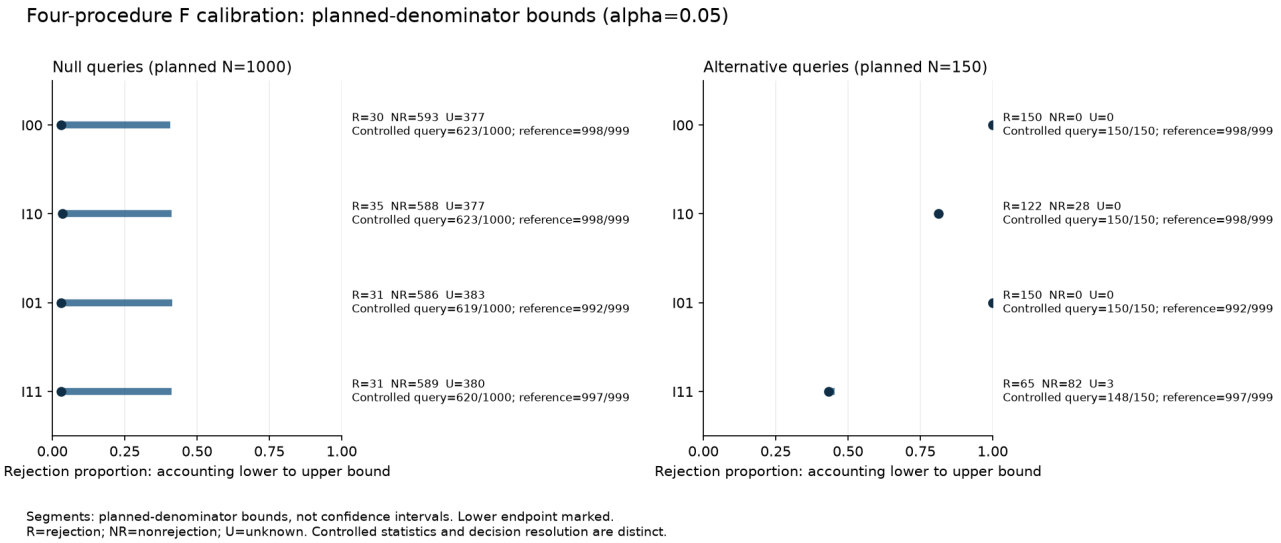


Figure 3: Actual final calibration on the full planned denominators. Segments span definite rejection through rejection plus unresolved outcomes; they are accounting bounds. The figure retains unknowns and controlled query/reference counts.